HANDS-ON 3 · DAY 1 / AFTERNOON · GROUP'S SHEET

Grading wines: from seven grades to three lines

The case. A winery sells its wines in three lines: low, medium and high. Today each wine waits for a tasting panel before it is placed. The winery asks whether its laboratory measurements could place a new wine first, and how far such a model can be trusted, class by class.

DATA CARD · WINEQUALITY_ALL.CSV

What a row is. One sample of vinho verde, red or white, from northern Portugal, tested at the region's official certification body between May 2004 and February 2007. 6,497 rows, 13 columns, no missing values.

The target. `quality`: the median of the scores of at least three blind tasters, from 0 (very bad) to 10 (excellent). Only grades 3 to 9 occur, and 5 and 6 hold 77% of the wines.

How the rows were chosen. Wines with a protected designation of origin, sent by their producers for certification. 4,898 are white, 1,599 red. Only lab measurements are given: no grape, brand or price.

Its history. The authors modelled red and white apart, as a regression of the grade. The course file joins their two files and adds `color`. 1,177 rows (18%) are exact copies of another row; the paper does not mention them. Panel 1 removes them.

Source and licence. P. Cortez, A. Cerdeira, F. Almeida, T. Matos and J. Reis (2009), Modeling wine preferences by data mining from physicochemical properties, *Decision Support Systems* 47(4):547 to 553. UCI Machine Learning Repository, dataset 186, doi:10.24432/C56S3T, CC BY 4.0.

column	what it measures	column	what it measures
`fixed acidity`	tartaric acid, g/dm³	`total sulfur dioxide`	mg/dm³, free and bound
`volatile acidity`	acetic acid, g/dm³	`density`	g/cm³
`citric acid`	g/dm³	`pH`	acidity on the pH scale
`residual sugar`	sugar left after fermentation, g/dm³	`sulphates`	potassium sulphate, g/dm³
`chlorides`	sodium chloride, g/dm³	`alcohol`	% of volume
`free sulfur dioxide`	mg/dm³	`color`	`red` or `white`
`quality`	**the target**: the grade, 0 to 10		

THREE ROLES, ONE SCREEN

Driver. Works Dataiku, and only Dataiku: says aloud what each screen shows before clicking on.

Mapper. Keeps this sheet: writes the counts, the scores and the answers.

Referee. Checks every step against this sheet and challenges every conclusion: is it true of the wines, or of the tool? Speaks for the group in the debrief.

STEP 1 · EXPLORE THE WINES, THEN REMOVE THE COPIES

1 ☐ **Load the data**
In a new project, on the Flow: **+ Dataset** → **Upload your files**, choose `winequality_all.csv`, then **Create**.

2 ☐ **Count the grades**
Open the dataset: **Explore**, click the header of `quality`, then **Analyze**; the same for `color`.

Wines graded 3: _____ graded 9: _____ share of grade 6: _____ white wines: _____

3 ☐ **Remove the copies**
Select the dataset, then **+ Recipe** → **Distinct**. Keep every column, **Create**, then **Run**.

Rows after **Distinct**: _____ copies removed: _____

STEP 2 · TRAIN ON THE SEVEN GRADES

1 ☐ **Create the analysis**
Select the **distinct** dataset, then **Lab** → **AutoML Prediction**. Target `quality`. Keep **Quick prototypes**, then **Create**.

2 ☐ **Say that the grades are classes**
Design → **Target**: if Dataiku proposes **Regression** (the grades are numbers), set the prediction type to **Multiclass classification**.

3 ☐ **Use five folds, then train**
Design → **Train/Test set**: **K-fold cross-test**, 5 folds, stratified. Click **Train**.

Metric **Results** ranks by: _____ best model: _____ score and ±: _____

STEP 3 · READ THE MODEL CLASS BY CLASS

1 ☐ **The confusion matrix**
Results → **model** → **Performance** → **Confusion matrix**. Rows are actual grades, columns predicted ones.

Grades never predicted: _____ share of the 6s predicted 6: _____ of the 7s predicted 7: _____

2 ☐ **One ROC curve per grade**
Performance → **ROC & PR curves**: choose the class 8.

AUC of grade 8: _____ How can a grade be ranked well and yet never be predicted?

☐ **Reached this line? Stop here: the Referee reports in the debrief.**

Panels 4 and 5 follow slide 14 of **Beyond binary classification**.

4 · THREE LINES INSTEAD OF SEVEN GRADES

1 ☐ **Regroup the grades**
On the distinct dataset: **+ Recipe** → **Prepare**, add a **Formula** step with output column `grade`:

```
if(quality <= 5, "low", if(quality == 6, "medium", "high"))
```

Run the recipe.

2 ☐ **Train on the lines**
On the prepared dataset: a new **Lab** → **AutoML Prediction**, target `grade`, the settings of panel 2. **Train**.

Best score: _____ All at 1.000? **Design** → **Features handling**: which column gives `grade` away? _____

3 ☐ **Remove the leak**
Reject that column, and **Train** again.

Best score now, and its ±: _____

5 · DECIDE FOR THE WINERY

target	accuracy	F1 (Dataiku's average)	recall, from the matrix
seven grades, panel 2			of 8:
three lines, panel 4			of high:

Design → **Metrics**: does Dataiku average the classes weighted by their size, or not? _____

Which version would the winery deploy, and which score should it watch?

Extensions. (a) Merge only the rare grades, 3 into 4 and 9 into 8: Formula `if(quality == 3, 4, if(quality == 9, 8, quality))`, multiclass again, `quality` rejected. Does the F1 rise? (b) **Design** → **Train/Test set**, **Sampling method**: **Class rebalance**. What happens to the rare grades' recall, and to accuracy? Which mix of wines does the score then describe?